Geographic Data Science

Space, formally

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Space, formally

Representing spatial context

A spatial statistical method needs to represent geography or spatial context.

Spatial weights matrices are one of the most common ways to do this.

Translating geography

(Geo)visualisation

Numbers into a visual language the human brain understands more easily.

Spatial weights matrices

Geography into a numerical language a computer understands.

Spatial analysis techniques

Spatial weights are a core element of:

- Spatial autocorrelation
- Spatial clustering
- Geodemographics
- Spatial regression

W as a formal representation of space

The spatial weights matrix, *W*

An $N \times N$ positive matrix recording spatial relationships between all observations in a sample.

By convention, $w_{ii} = 0$.

What is a neighbour?

Types of *W*

What is a neighbour?

A neighbour is “somebody” who is:

Next door	Contiguity-based weights
Close	Distance-based weights
In the same “place”	Block weights

See Anselin & Rey (2014) for a detailed discussion and further types of W.

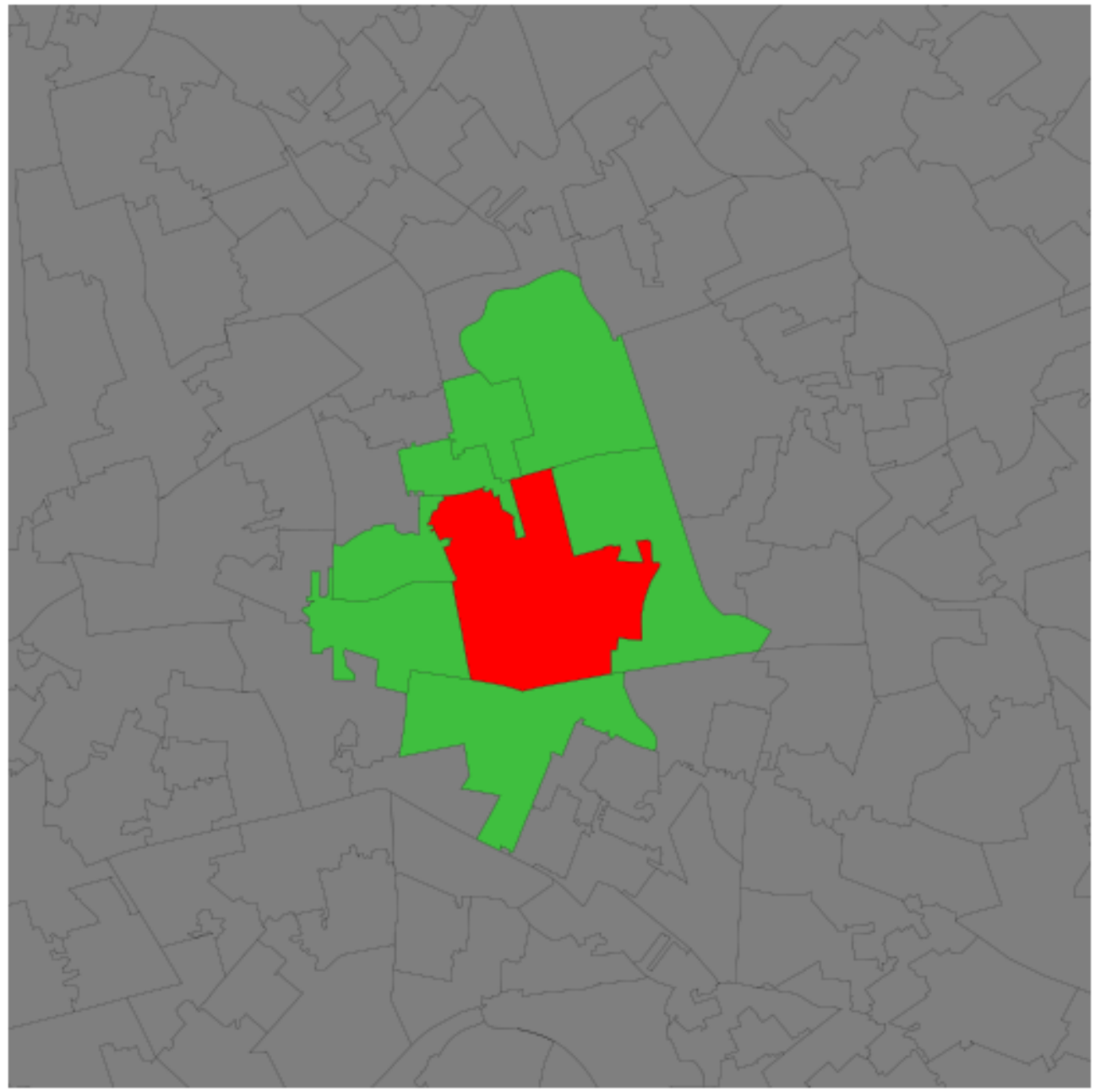
Contiguity-based weights

Neighbours share boundaries to some extent.

- Rook
- Queen

Other contiguity definitions are possible.

Contiguity: map example



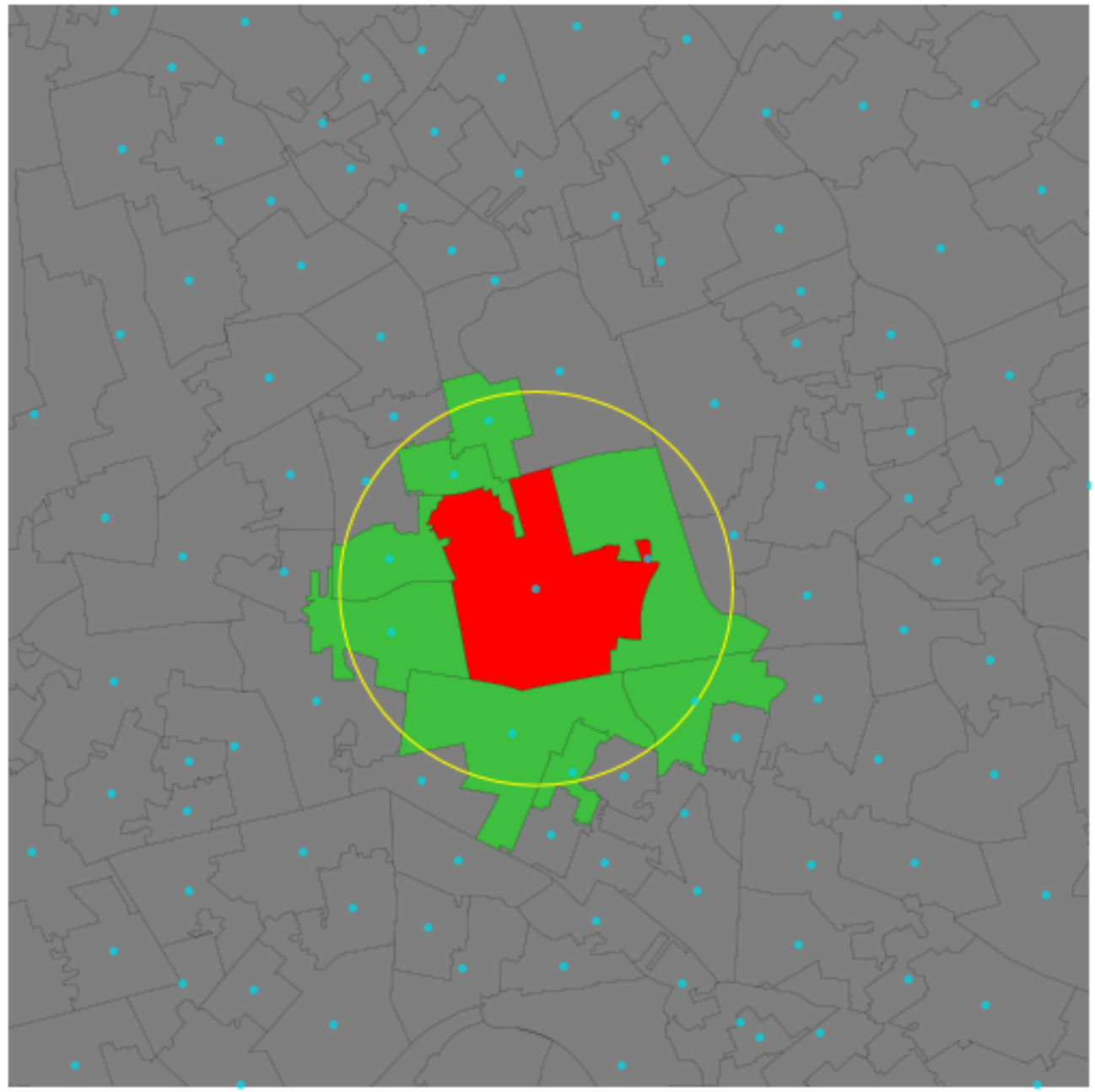
Distance-based weights

Weights relate to the distance between observations, often inversely.

- Inverse distance, with a distance threshold
- *k*-nearest neighbours (KNN), with a fixed number of neighbours

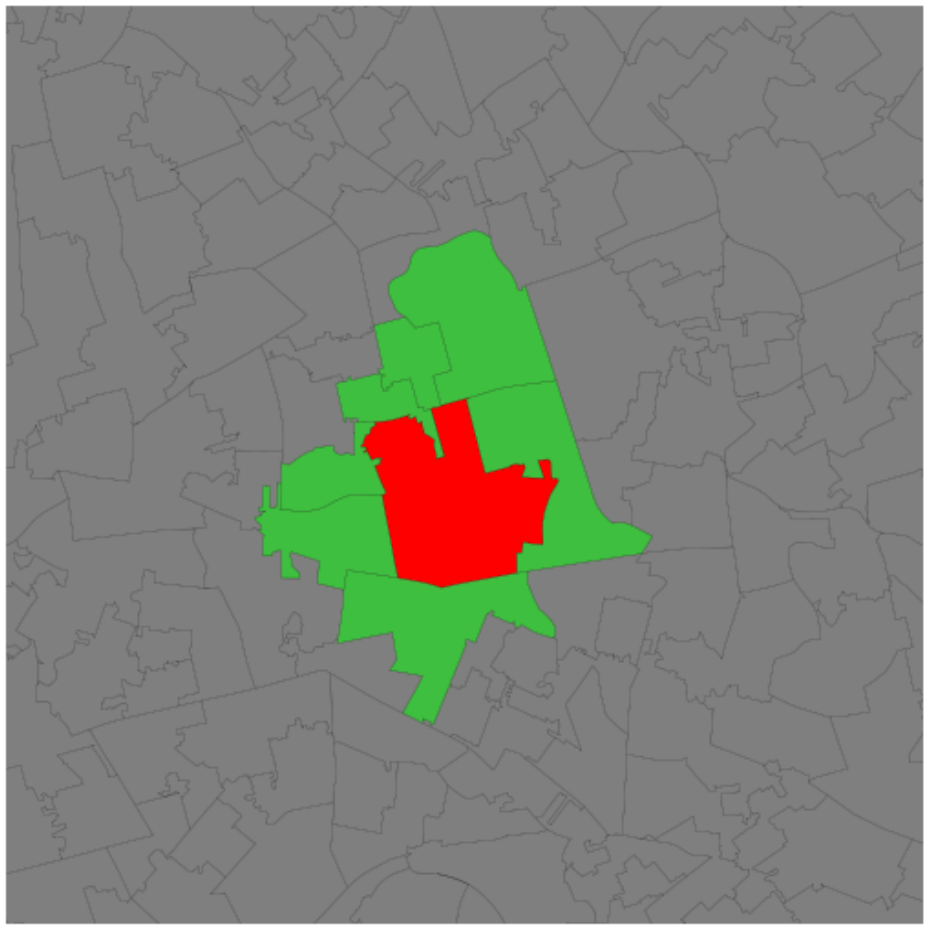
Other distance definitions are possible.

Distance: map example

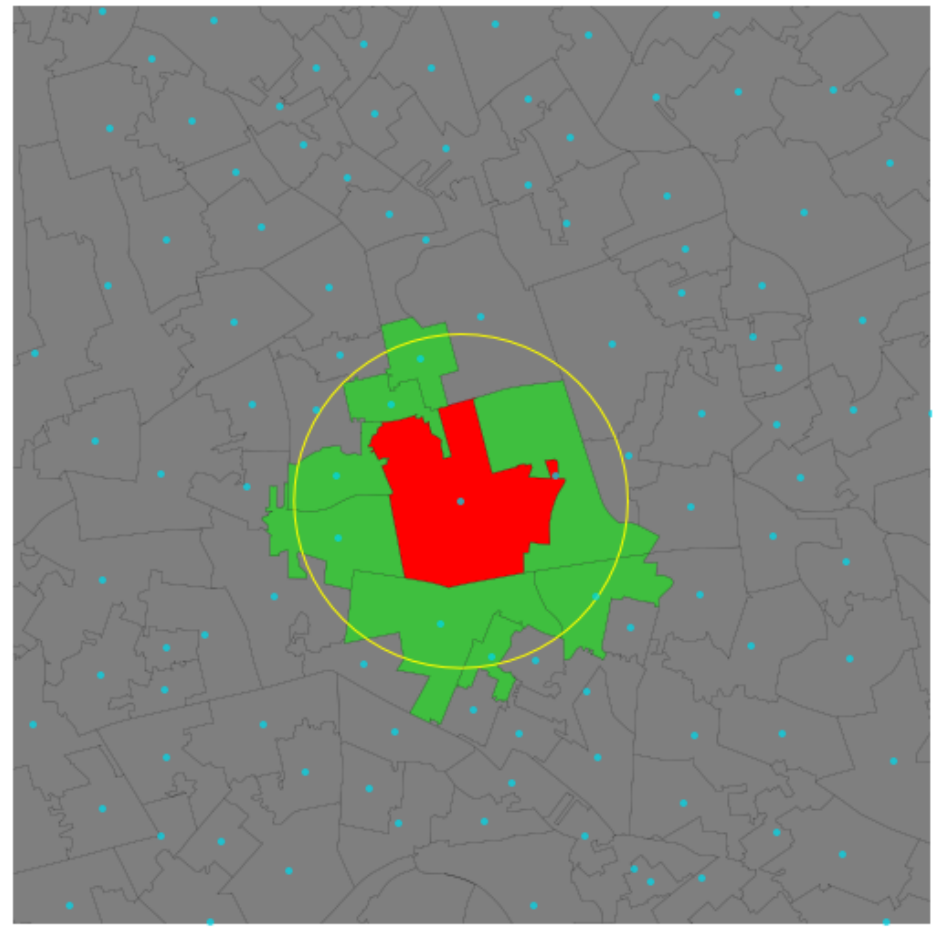


Contiguity and distance

Queen neighbours



Neighbours within 1 km



Block weights

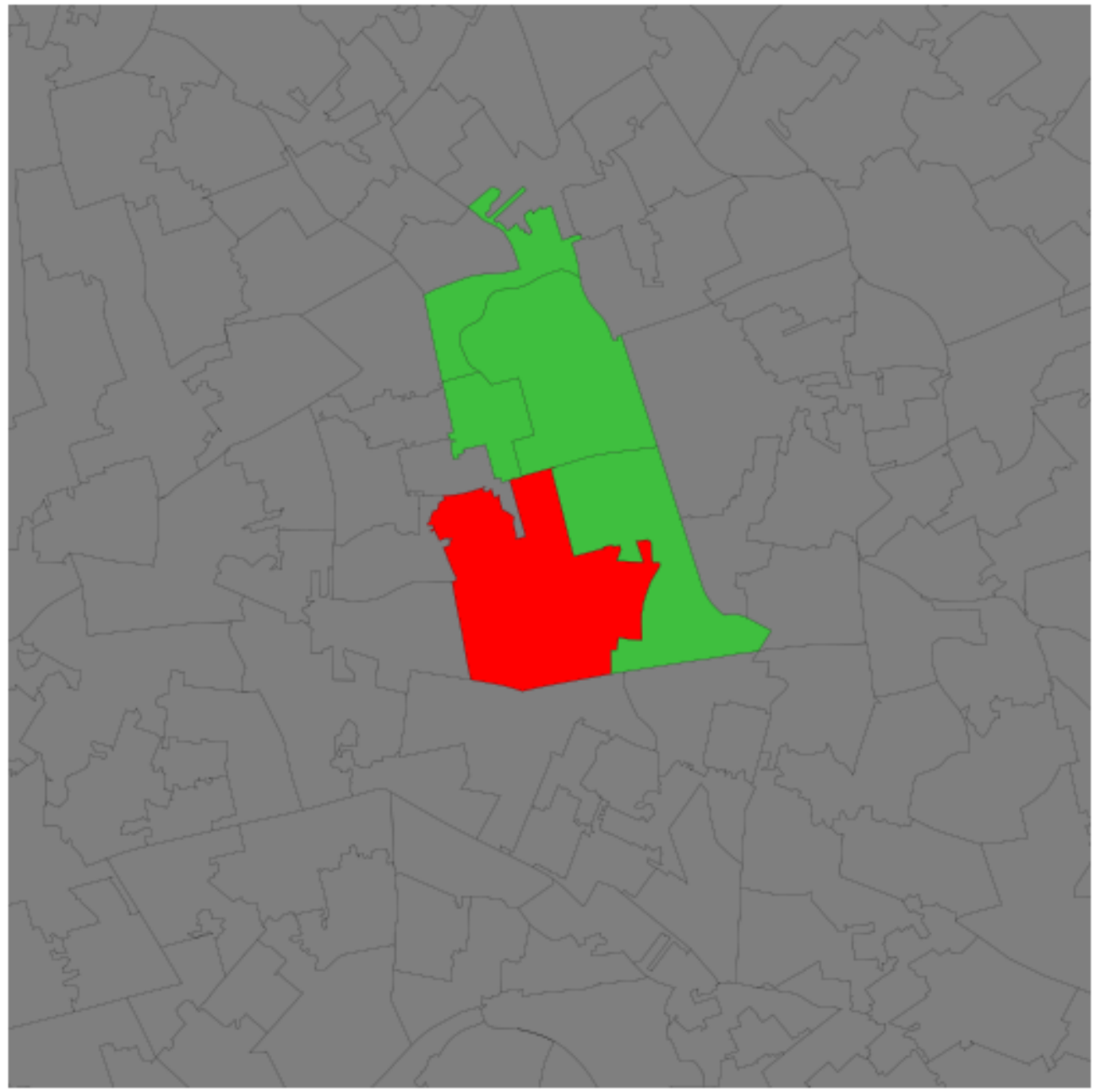
Weights follow discretionary rules loosely related to geography.

Examples

- LSOAs grouped into MSOAs
- Postcodes within city boundaries
- Counties within states

Other groupings are possible.

Block weights: map example



How much of a neighbour?

Non-neighbours receive zero weight:

$w_{ij} = 0$

For neighbours, w_{ij} depends on the chosen weighting rule.

Choice of *W*

W should reflect the channels of interaction relevant to the question.

Contiguity weights	Immediate contact, e.g. disease contagion
Distance weights	Accessibility
Block weights	County differences in laws

Standardisation

Some applications, including spatial autocorrelation, commonly standardise *W*.

Row standardisation divides each element by the sum of its row.

$$w_{ij}^{*} = w_{ij} / w_i$$

$$w_i = \sum_j w_{ij}$$

w_i is the sum of row *i*

Geographic Data Science

The spatial lag

The spatial lag

The product of a spatial weights matrix *W* and a variable *Y*.

$$Y_{sl} = WY$$

$$y_{sl,i} = \sum_j w_{ij} y_j$$

Interpreting the spatial lag

It captures the behaviour of a variable in the neighbourhood of observation *i*.

With row-standardised *W*, it is the average value of the variable in that neighbourhood.

Using the spatial lag

A common way to introduce space into a statistical framework.

Used in ESDA and spatial regression to delineate neighbourhoods, including:

- Moran's *I*
- LISAs
- Spatial models (lag, error, …)

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Dani Arribas-Bel for a course in Geographic Data Science.

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